

# BERT Shows Garden Path Effects

Kyra Wilson<sub>1</sub>, Tovah Irwin<sub>1</sub>, Alec Marantz<sub>1, 2</sub>

<sup>1</sup>New York University Abu Dhabi | <sup>2</sup>New York University

## INTRODUCTION

?? The horse raced past the barn fell. ??

- Sequential processing leads to **temporary ambiguity**
- Syntactic parsing and semantic role assignment is more complicated** because of the ambiguity
- Bidirectional language models might be superior to humans at garden path processing** because they receive all tokens simultaneously

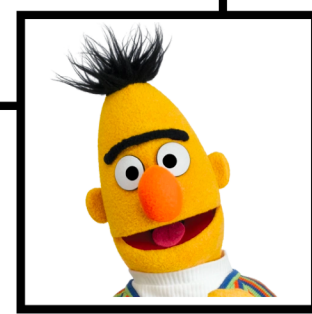
## QUESTION ANSWERING

|                 |                                                                                  |
|-----------------|----------------------------------------------------------------------------------|
| Garden Path     | The child ( <u>who was</u> ) <u>bought the ice cream cone</u> hugged the nanny.  |
| Local Coherence | The nanny hugged the child ( <u>who was</u> ) <u>bought the ice cream cone</u> . |

### BERT Experiment

- 160 contexts x 5 question types = 800 context + question pairs
- Pre-trained models from BERT family, fine-tuned for QA using SQuAD2.0

Context: The nanny hugged the child bought the ice cream cone.  
Question: Who hugged the child?

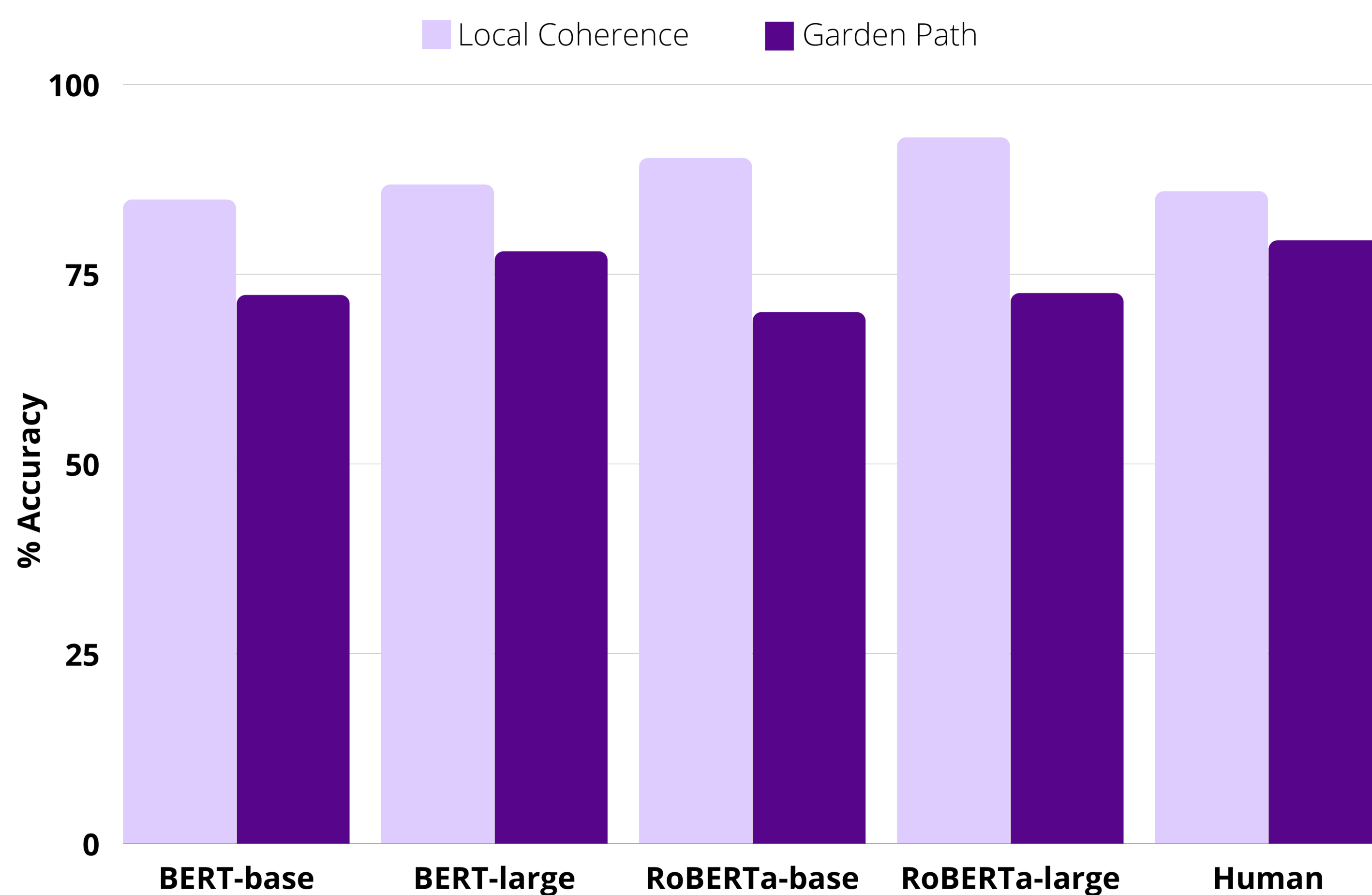


### Behavioral Experiment

- 74 English speakers recruited from Prolific
- 45 random items: 32 target + 13 filler
- > 80 responses for each of the sentence and question types

The nanny hugged the child bought the ice cream cone.

Who hugged the child?



## REFERENCES

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## OBJECTIVES

Do humans and BERT perform differently in answering questions about garden path sentences given their distinct processing strategies?

Does BERT also struggle to assign semantic roles in garden path sentences, akin to humans?

## PROBE TASK

### Probe Training

We trained binary linear classifiers on our model's word embeddings across all layers to probe representations of the **Agent** and **Goal** roles, using annotated sentences from PropBank [5]. These roles were used most frequently by PropBank frames for verbs in our stimuli set.



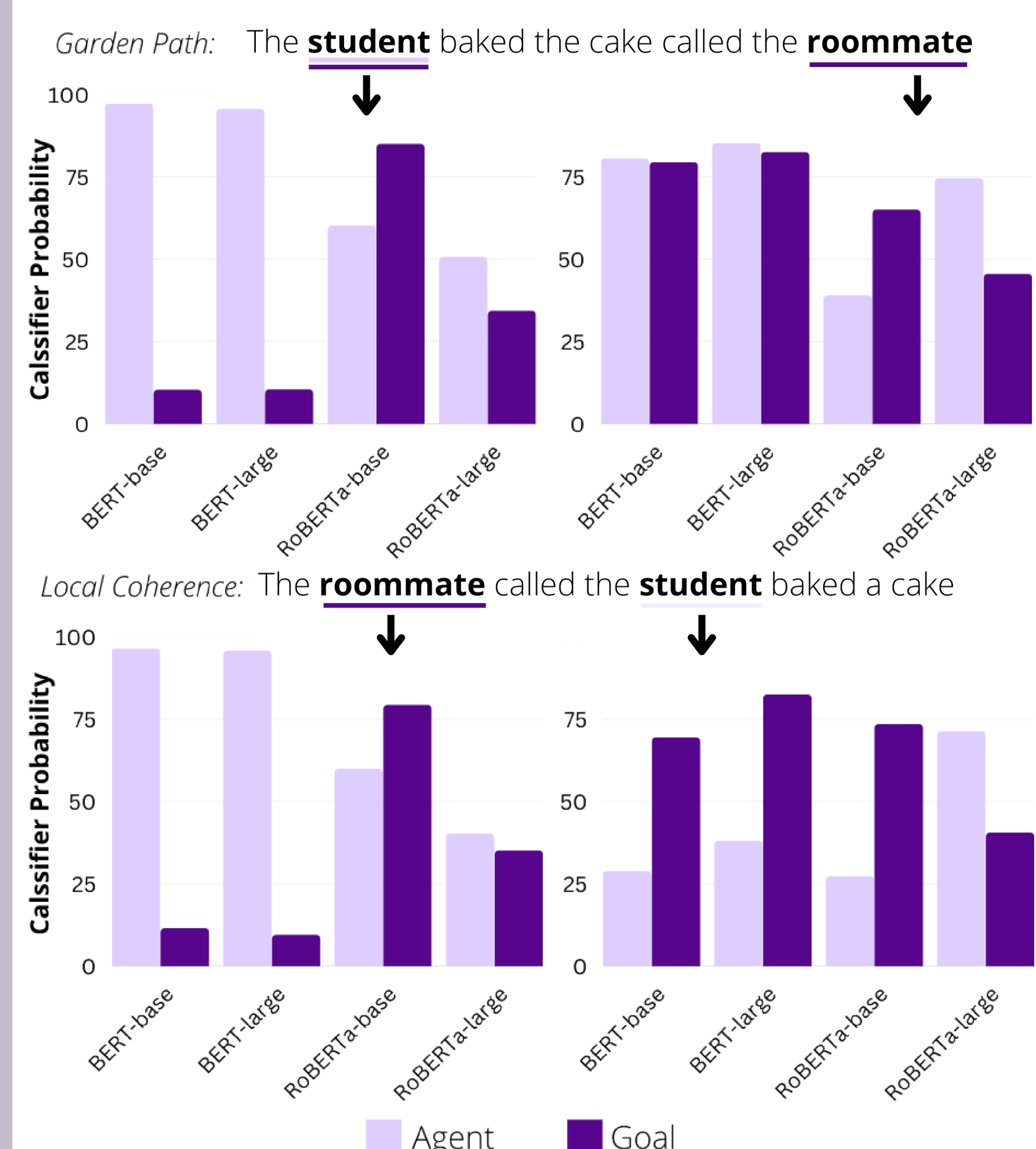
- 80/20 train test split
- High accuracy on Propbank test sentences

| Model         | Layer | Agent Accuracy | Goal Accuracy |
|---------------|-------|----------------|---------------|
| BERT-base     | 9     | 84.92          | 75.96         |
| BERT-large    | 17    | 84.44          | 75.12         |
| RoBERTa-base  | 11    | 64.2           | 62.6          |
| RoBERTa-Large | 10    | 66.8           | 66.44         |

Probe Results on Propbank sentences for layers with highest accuracy and selectivity

### Probing Garden Paths

- Tested only sentences containing both Agent and Goal roles, according to PropBank (97/160 sentences total)



- The **BERT models fail to assign the Goal role** to the head of the embedded clause
- RoBERTa models classify Goal** more frequently
- Size doesn't matter!** Larger models do not perform better universally
- Role classifiers perform very differently** between BERT and RoBERTa models despite identical probe hyperparameters and training set

## CONCLUSION

- BERT and RoBERTa models perform similarly** on question answering task, regardless of size
- BERT family models are worse than humans** on garden path sentences despite a linear input
- These results **may be due to a failure to assign the correct semantic role in relative clauses**, following theories from human sentence processing
- The models' are inconsistent** in which roles are misassigned

Check out our paper and code!



[tinyurl.com/gardenBERT](https://tinyurl.com/gardenBERT)